

Notes: Rauh (JF, 2006)

Investment and Financing Constraints: Evidence from the Funding of Corporate Pension Plans

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Defined-benefit pension funding	2
2.2	Contribution rules	3
2.3	Sample construction	3
2.4	Main variables	3
3	Models and empirical specifications	3
3.1	Financing-constraint prediction	3
3.2	Standard investment-cash flow regression	4
3.3	Main mandatory-contribution regression	4
3.4	Funding-status controls	4
3.5	Predictable and unexpected components	4
3.6	Industry offset regression	4
4	Identification and empirical design	5
4.1	What identifies the main effect	5
4.2	Why pension shocks are useful	5
4.3	What the paper cannot fully prove	5
5	Tables and figures	5
5.1	Figures 1–2: funding status and mandatory contribution rules	6
5.2	Table I and Appendix Table AI: sample and summary statistics	6
5.3	Figures 3–5: contribution estimates and nonparametric evidence	7
5.4	Tables II–III: main capital-expenditure results and alternative specifications	8
5.5	Tables IV–V: other uses of funds and predictable vs. unexpected MCs	9
5.6	Table VI and Figure 6: financing-constraint heterogeneity and timing	10
5.7	Table VII: investment response of non-DB firms	11
6	Short summary	12
7	Conclusion	12
7.1	Core conclusion	12
7.2	Economic intuition	13
7.3	Takeaway	13

1 Big picture

- **Question.** Do shocks to internal financial resources reduce corporate investment when firms face financing constraints?
- **Identification idea.** The paper uses mandatory contributions to underfunded defined-benefit pension plans as cash drains. The contribution rules are sharply nonlinear in pension funding status, creating plausibly exogenous variation in internal funds.
- **Institutional shock.** Under ERISA and later pension laws, firms with underfunded plans must contribute cash according to legal formulas. Firms with overfunded plans face no comparable mandatory drain.
- **Core empirical strategy.** Estimate the effect of mandatory contributions on capital expenditures while controlling for Tobin's Q, cash flow, firm and year fixed effects, and the pension funding status itself.
- **Main result.** Capital expenditures fall by roughly \$0.60–\$0.70 for each \$1 of mandatory pension contributions in the preferred specifications; some estimates are closer to \$0.80.
- **Why this matters.** Standard investment-cash flow regressions often have coefficients around 0.10. The pension contribution shock implies a much larger marginal effect of internal funds when the cash flow variation is better identified.
- **Financing constraints.** The effect is strongest for firms that look financially constrained: firms without strong credit ratings, younger firms, low-dividend firms, low-cash firms, and firms whose capital expenditures often exceed cash flow.
- **Unexpected component.** Investment also falls with the component of mandatory contributions caused by unexpected asset-market and interest-rate movements.
- **Other uses of funds.** Mandatory contributions appear to reduce some acquisitions, dividends, and repurchases, but the strongest and most robust effect is on capital expenditures.
- **General equilibrium offset.** Non-DB firms in the same industry increase investment when pension firms face high contribution requirements, offsetting about 12 percent of the lost DB-firm investment.
- **Economic takeaway.** Required pension funding creates a clean internal-cash shock. The results support the view that financing frictions make investment depend on internal resources.

2 Data and measurement

2.1 Defined-benefit pension funding

- A defined-benefit pension plan promises retirees benefits based on salary, tenure, and age.
- The plan is *overfunded* if pension assets exceed pension liabilities and *underfunded* if assets are below liabilities.
- U.S. law requires firms with underfunded plans to contribute cash to pension funds.
- Rauh uses plan-level IRS Form 5500 data because Compustat pension disclosures are not sufficient to calculate legal contribution requirements in the sample period.

Interpretation. Mandatory pension contributions are cash outflows that are not chosen freely by the firm at the time of investment. They therefore provide a useful instrument-like source of variation in internal funds.

2.2 Contribution rules

- During the sample, firms were required to contribute the larger of the minimum funding contribution (MFC) and the deficit reduction contribution (DRC).
- The MFC is approximately normal cost plus an amortization of ERISA underfunding.
- The DRC is a nonlinear catch-up rule that depends sharply on the degree of underfunding.
- The required contribution falls to zero at full funding, creating a discontinuity/kink around zero funding status.

Interpretation. The nonlinear legal schedule is the core source of identification. Investment opportunities may vary smoothly with funding status, but they are unlikely to have exactly the same kinks and asymmetries as the legal contribution rule.

2.3 Sample construction

- The main sample covers 1990–1998, the period for which the Department of Labor IRS 5500 data are available.
- The final sample contains 8,030 firm-year observations and 1,522 firms after matching Compustat and pension data and requiring variables needed for investment regressions.
- The sample includes firms with U.S. defined-benefit pension plans with more than 100 participants.
- Firm-level pension variables are created by aggregating plan-level data to the firm-year level.

Interpretation. The empirical sample is large enough for panel regressions but still targeted to firms for which legal pension contribution requirements can be constructed.

2.4 Main variables

- **Capital expenditures:** capital expenditures scaled by beginning-of-year book assets.
- **Mandatory contributions:** estimated legal required pension contributions scaled by beginning-of-year assets.
- **Funding status:** overfunding minus underfunding, scaled by assets.
- **Cash flow:** net income plus depreciation and amortization plus pension expense minus actual pension contributions.
- **Nonpension cash flow:** net income plus depreciation and amortization plus pension expense, excluding the actual pension contribution.
- **Tobin’s Q:** market-to-book asset ratio.

Interpretation. The distinction between cash flow and nonpension cash flow matters because pension contributions are cash outflows that are not always deducted in standard cash-flow measures.

3 Models and empirical specifications

3.1 Financing-constraint prediction

Kaplan and Zingales’ two-period model implies that when external finance is costly,

$$\frac{dI^*}{dw} = \frac{-C_{11}}{C_{11} - f_{11}}, \quad (1)$$

where w is internal wealth, $f(I)$ is the return to investment, and $C(e, \theta)$ is the cost of external finance.

Interpretation. If mandatory pension contributions reduce internal wealth, investment should fall for constrained firms. The magnitude depends on the curvature of investment returns and external-finance costs.

3.2 Standard investment-cash flow regression

The conventional specification is

$$\frac{I_{it}}{A_{i,t-1}} = \alpha_i + \alpha_t + \beta_1 Q_{i,t-1} + \beta_2 \frac{CF_{it}}{A_{i,t-1}} + \varepsilon_{it}. \quad (2)$$

Interpretation. This regression is hard to interpret because cash flow may proxy for unobserved investment opportunities or measurement error in Q .

3.3 Main mandatory-contribution regression

Rauh estimates

$$\frac{I_{it}}{A_{i,t-1}} = \alpha_i + \alpha_t + \beta_1 Q_{i,t-1} + \beta_2 \frac{NonPensionCashFlow_{it}}{A_{i,t-1}} + \beta_3 \frac{Z_{it}}{A_{i,t-1}} + \mathbf{x}_{it}\gamma + \varepsilon_{it}, \quad (3)$$

where Z_{it} is actual total contributions or estimated mandatory contributions.

Interpretation. The coefficient β_3 is the main object. A negative coefficient means legally required pension contributions crowd out investment.

3.4 Funding-status controls

The paper adds increasingly flexible controls for pension funding status:

- linear funding status;
- separate underfunding and overfunding terms;
- polynomial controls in underfunding and overfunding.

Interpretation. These controls address the possibility that pension funding status is correlated with investment opportunities. The identifying variation comes from the legal contribution function conditional on funding status.

3.5 Predictable and unexpected components

Mandatory contributions are decomposed into predictable and unexpected parts:

$$MC_{it} = MC_{it}^{pred} + MC_{it}^{unexp}. \quad (4)$$

The unexpected component is driven by deviations of pension asset returns and interest rates from expected values.

Interpretation. If investment responds to the unexpected component, the result is difficult to attribute to planned investment opportunities or slow-moving firm fundamentals.

3.6 Industry offset regression

For firms without defined-benefit pension assets, Rauh estimates

$$\frac{CAPX_{it}}{A_{i,t-1}} = \alpha_i + \alpha_t + \beta_1 Q_{i,t-1} + \beta_2 \frac{CashFlow_{it}}{A_{i,t-1}} + \beta_3 IndustryPensionRequirement_{h(i),t} + \varepsilon_{it}. \quad (5)$$

Interpretation. This asks whether nonpension firms in the same industry take up investment opportunities that constrained pension sponsors forgo.

4 Identification and empirical design

4.1 What identifies the main effect

- The legal contribution function is sharply nonlinear and asymmetric around underfunding thresholds.
- The regressions control directly for the pension funding status itself.
- Firm fixed effects absorb time-invariant firm characteristics.
- Year fixed effects absorb aggregate macroeconomic and investment shocks.
- Tobin's Q and nonpension cash flow control for standard investment determinants.

Interpretation. The key identifying assumption is that unobserved investment opportunities may depend on funding status, but not with precisely the same legal kinks and jumps that determine mandatory contributions.

4.2 Why pension shocks are useful

- Pension assets are invested in financial markets; market movements affect plan funding.
- Pension liabilities depend on discount rates, especially the 30-year Treasury rate in the sample period.
- These movements change required contributions without necessarily reflecting the firm's product-market investment opportunities.

Interpretation. This separates the cash-flow shock from the standard omitted-opportunities problem in investment-cash flow regressions.

4.3 What the paper cannot fully prove

- Pension funding status itself can be linked to firm health, even with controls.
- Mandatory contribution estimates require actuarial and regulatory approximations.
- The data end before the large 2001–2003 pension underfunding episode.
- Some investment lost by DB firms may be shifted to non-DB firms or to later years.
- The paper identifies effects for defined-benefit pension sponsors, not necessarily all firms.

Interpretation. The design is not a randomized experiment, but the nonlinear funding-rule variation is much cleaner than ordinary cash-flow variation.

5 Tables and figures

This section keeps the Read plus Economic message format and groups adjacent figures/tables to save space. Images are directly cropped from the paper.

5.1 Figures 1–2: funding status and mandatory contribution rules

Read:

- Figure 1 shows that pension funding status varies substantially over time. The distribution deteriorates sharply after 2000.
- Figure 2 shows the legal contribution schedule. Mandatory contributions are positive only for underfunded plans and are highly nonlinear in the degree of underfunding.
- The DRC rule becomes more demanding after the 1994 Retirement Protection Act.

Economic message. The institutional setting creates cash drains that are large, nonlinear, and tied to pension funding rather than directly to investment demand.

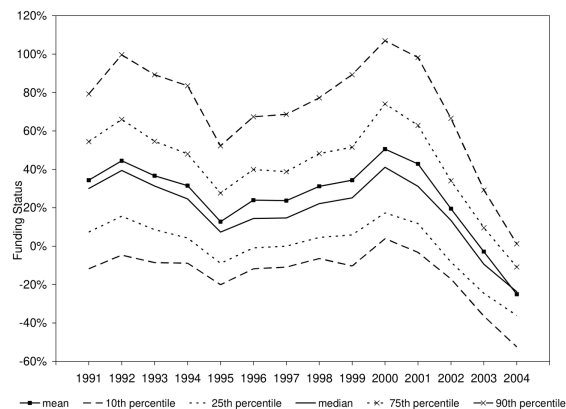


Figure 1. Distribution of beginning-of-year funding status. This figure shows the distribution of the firm-level pension funding status as of the start of the fiscal year for Compustat firms during 1991 to 2004. The funding status is defined as pension assets minus pension liabilities divided by pension liabilities. The data are from the annual filings of companies in the Compustat

(a) Funding status distribution

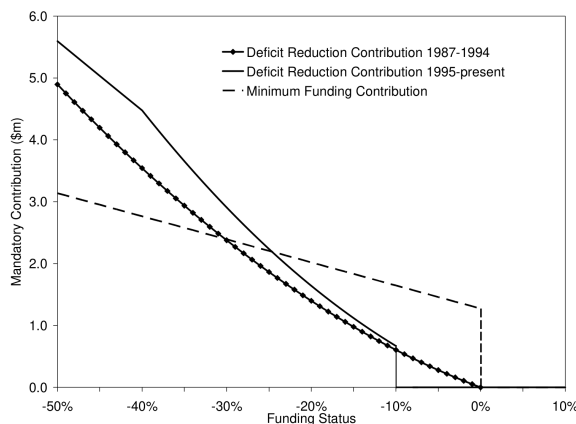


Figure 2. Mandatory pension contributions. A firm's required pension contribution is the

(b) Mandatory contribution schedule

5.2 Table I and Appendix Table AI: sample and summary statistics

Read:

- Table I reports 8,030 firm-year observations and 1,522 firms in the main sample.
- Mean capital expenditures are 6.9 percent of assets and median capital expenditures are 5.8 percent.
- Mandatory contributions are positive for 2,380 observations; conditional on being nonzero, the 90th percentile is 0.8 percent of assets and 28.9 percent of lagged capital expenditures.
- Appendix Table AI shows how the sample is constructed from Compustat firms to DB pension firms matched to IRS 5500 filings.

Economic message. The data contain many small required contributions, but the upper tail is economically meaningful relative to investment.

Table I Summary Statistics									
This table presents summary statistics for the main sample of 8,030 firm-year observations. For inclusion in the sample, a firm-year observation must be found in both Compustat and the IRS 5500 database of U.S. defined benefit plans with more than 100 participants (see Appendix Table I for sample construction). All variables are winsorized at the 1 st and 99 th percentiles of their distributions. Variables subscripted with ₋₁ are the values as of the beginning of year. Most variables are as of fiscal year-end, scaled by beginning-of-year balance sheet assets at book value (data6). <i>Capital Expenditures</i> are from the firm's statement of cash flows (data128). <i>Cash Flow</i> equals nonpension cash flow minus total pension contributions. <i>Nonpension Cash Flow</i> equals net income plus depreciation and amortization plus the accounting definition of the pension expense (data14 + data18 + data43); this is essentially the same measure as used by Kaplan and Zingales (1997) and Baker, Stein, and Wurgler (2003), but the accounting definition of pension expense is added back as a noncash component of reported earnings. <i>Total Pension Contributions</i> are reported on the IRS 5500 forms at the plan level and in this study are aggregated to the firm level. <i>Total U.S. Pension Assets</i> are the current value of pension assets as of the beginning of the year from the IRS 5500s. <i>Total U.S. Pension Liabilities (ABO)</i> are accumulated benefit obligation current liabilities from Schedule B of the IRS 5500s. For years 1991 to 1994, this variable is the OBRAS7 liability and for years 1995 to 1998, it is the RPA94 liability. <i>Underfunding (Overfunding)</i> equals the market value of assets in underfunded (overfunded) plans minus the current liabilities of underfunded plans. <i>Funding Status</i> is <i>Overfunding - Underfunding</i> , or alternatively, total pension assets minus total pension liabilities. <i>Tobin's Q</i> is the market value of equity (data199 + data25) plus book assets (data6) minus the book value of common equity including deferred taxes (data80 + data74) over assets, as in Baker, Stein, and Wurgler (2003). <i>Mandatory Contributions</i> are the estimated mandatory component of contributions as described in Section II.									
Years 1990 to 1998, Observations = 8,030, Firms = 1,532									
	Mean	Median	Standard Deviation	Nonzero Observations	Percentiles, Conditional on Nonzero				
					10 th	25 th	50 th	75 th	90 th
Capital expenditures/assets ₋₁	0.069	0.058	0.053	8,030	0.018	0.034	0.058	0.089	0.131
Cash flow/assets ₋₁	0.096	0.096	0.077	8,030	0.014	0.058	0.096	0.140	0.182
Nonpension cash flow/assets ₋₁	0.099	0.099	0.077	8,030	0.017	0.061	0.099	0.143	0.187
Total pension contributions/assets ₋₁	0.003	0.001	0.006	5,562	0.000	0.001	0.003	0.007	0.013
Pension expense/assets ₋₁	0.007	0.005	0.008	8,006	0.000	0.002	0.005	0.010	0.017
Total U.S. pension assets/assets ₋₁	0.137	0.087	0.167	8,030	0.010	0.033	0.087	0.181	0.329
Total U.S. pension liabilities (abo)/assets ₋₁	0.112	0.067	0.134	8,030	0.009	0.028	0.067	0.140	0.270
Total global pension assets/assets ₋₁	0.172	0.117	0.174	7,766	0.024	0.060	0.121	0.235	0.409
Underfunding/assets ₋₁	0.004	0.000	0.014	3,021	0.000	0.001	0.003	0.011	0.032
Overfunding/assets ₋₁	0.030	0.011	0.046	6,624	0.001	0.005	0.018	0.046	0.095
Funding Status/assets ₋₁	0.025	0.010	0.050	8,029	-0.007	0.000	0.010	0.036	0.082
Tobin's Q (beginning of year)	1.479	1.256	0.707	8,030	0.901	1.034	1.256	1.681	2.329
Firm age (years)	26.3	28.0	14.1	8,030	5	13	28	39	44
Assets (\$m)	3,632	737	8,621	8,030	76	209	737	2697	9,220
Mandatory contributions/assets ₋₁	0.003	0.000	0.003	2,380	0.000	0.000	0.001	0.003	0.008
Mandatory contributions/cash flow ₋₁	0.009	0.000	0.043	2,380	0.000	0.001	0.008	0.031	0.099
Mandatory contributions/capital expenditures ₋₁	0.033	0.000	0.132	2,380	0.000	0.004	0.199	0.084	0.289

Figure 2: Table I: Summary statistics

Table AI Composition of Sample						
This table contains observation counts by fiscal year for the sample at different stages of construction. The starting sample (1) is all Compustat firms with a reported level of capital expenditures. Columns (2.1)–(2.3) show the sample counts for the firms that never have DB pension assets, for firms that have DB assets in at least 1 year, and for firms that have DB assets in the given year, respectively. Column (3) is the sample that matches with the IRS 5,500 data set. The number in brackets in column (3) is the number of plan observations from the tax filings that were collapsed to obtain the number of firm observations in each row of column (3). The match was done sequentially by CUSIP, name, and EIN. The majority of DB plans are sponsored by companies that are not publicly traded and hence a large part of the IRS 5,500 database is not used in this study. Firms with multiple plans are retained in the IRS sample. However, if a firm has multiple plans with different fiscal year-end months, all the firm's plans are dropped from the sample. Requirements for the final sample are: pension expense on an accounting basis is reported in Compustat; pension fiscal year must match firm fiscal year; requisite Compustat data must exist to compute capital expenditures, <i>Q</i> , cash flow, and assets; and IRS 5500 filing is complete enough to determine the funding status.						
Year	Start (1)	Non-DB vs. DB Pension Samples			Match (3)/(3-Plan)	Finish (4)
		(2.1)	(2.2)	(2.3)		
1990	6,421	4,265	2,156	1,943	1,240 [2,810]	746
1991	6,483	4,311	2,172	1,942	1,619 [3,380]	947
1992	6,662	4,462	2,200	1,963	1,499 [2,951]	883
1993	7,079	4,849	2,230	1,984	1,612 [3,103]	997
1994	7,474	5,201	2,273	2,019	1,660 [3,188]	1,031
1995	7,684	5,419	2,265	1,972	1,009 [1,866]	660
1996	8,403	6,080	2,323	2,017	1,355 [2,562]	918
1997	8,511	6,247	2,264	1,964	1,394 [2,431]	919
1998	8,151	6,014	2,137	1,876	1,446 [2,588]	949
	66,868	46,848	20,020	17,680	12,834 [24,879]	8,050
Capital expenditures in Compustat	•	•	•	•	•	•
Never has DB	•	•				
Has DB plan any year	•		•			
that year	•			•	•	•
Matched IRS					•	•
All variables						•

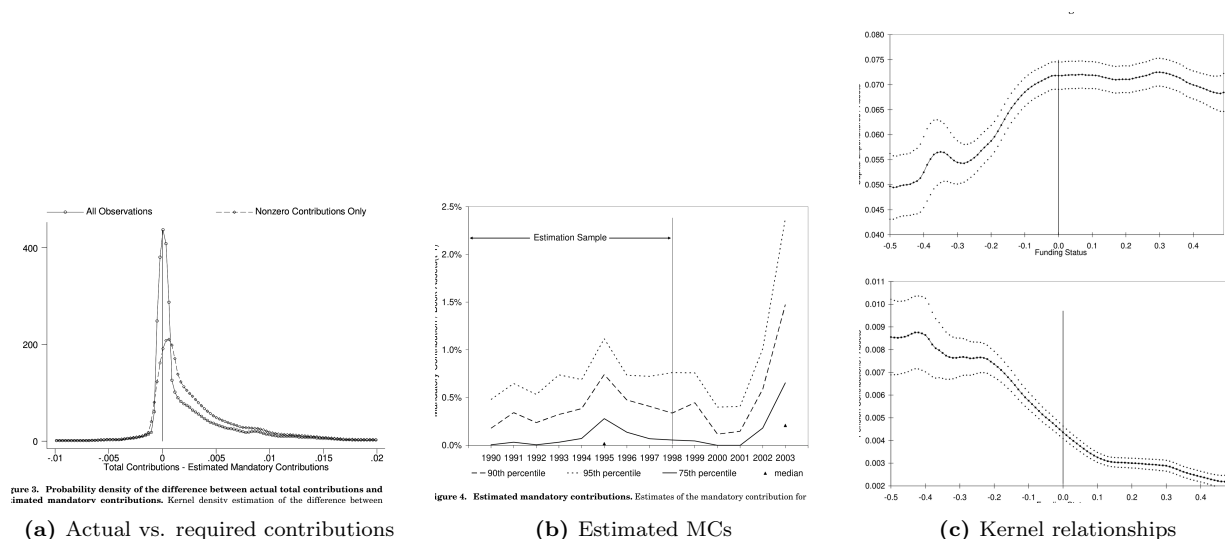
Figure 3: Appendix Table AI: Composition of sample

5.3 Figures 3–5: contribution estimates and nonparametric evidence

Read:

- Figure 3 shows that actual total contributions cluster around estimated mandatory contributions.
- Figure 4 shows that estimated mandatory contributions were modest in 1990–1998 but become large in simulated 1999–2003 calculations.
- Figure 5 shows capital expenditures rising with funding status up to full funding and pension contributions falling as funding status improves.

Economic message. The figures motivate the linear regressions: poor funding generates required cash drains, and investment appears lower near severe underfunding.



5.4 Tables II–III: main capital-expenditure results and alternative specifications

Read:

- Table II shows a coefficient of -0.830 on mandatory contributions without funding status controls in column (1c).
- With linear funding-status controls, column (2c) estimates -0.738.
- With separate underfunding/overfunding and polynomial controls, the estimates remain negative at about -0.607 and -0.597.
- Table III shows robustness to 2SLS, OLS, random effects, and first differences. Mandatory contribution coefficients remain negative.

Economic message. A dollar of mandatory pension contributions reduces capital expenditures by roughly 60–80 cents in the central specifications.

Table II

Panel Regressions of Capital Expenditures on Pension and Nonpension Cash Flows

Each column presents estimates from a regression of the form:

$$\frac{I_{i,t}}{A_{i,t-1}} = \alpha_0 + \alpha_1 Q_{i,t-1} + \beta_1 \frac{NonPensionCashFlow_{i,t}}{A_{i,t-1}} + \beta_2 \frac{Z_{i,t}}{A_{i,t-1}} + \mathbf{x}_{i,t}'\gamma + \epsilon_{i,t}$$

where $I_{i,t}$ is capital expenditures, $Z_{i,t}$ is (mandatory) pension contributions, and $\mathbf{x}$ is a vector of controls. Variables are scaled by beginning-of-year balance sheet asset ($A_{i,t-1}$). In specifications (1a), (2a), and (3a), nonpension cash flow and contributions are aggregated into one cash flow variable. In (1b), (2b), and (3b), the contributive variable is total contributions; in (1c), (2c), (3b), and (3c), it is mandatory contributions. In specifications (2a)–(2c), the funding status is controlled for linearity by including the funding status (pension assets minus pension liabilities scaled by firm assets) as an explanatory variable. In the specifications (3a)–(3b) the funding variable controls are underfunding and overfunding, separately, and in (3c) the first three powers of these variables are included (squares and cubes not shown). The sample size is 8,030 observations on 1,522 firms.

	Dependent Variable: Capital Expenditures _{<i>i,t</i>} / A _{<i>i,t-1</i>}								
	(1a)	(1b)	(1c)	(2a)	(2b)	(2c)	(3a)	(3b)	(3c)
Contributions (mandatory) _{<i>i,t</i>} / A _{<i>i,t-1</i>}			-0.830*** (0.289)			-0.738*** (0.284)		-0.607*** (0.296)	-0.597*** (0.300)
Contributions (total) _{<i>i,t</i>} / A _{<i>i,t-1</i>}		0.109 (0.162)			0.188 (0.158)				
Cash flow _{<i>i,t</i>} / A _{<i>i,t-1</i>}	0.111*** (0.012)								
Nonpension cash flow _{<i>i,t</i>} / A _{<i>i,t-1</i>}		0.111*** (0.012)	0.112*** (0.012)	0.111*** (0.012)	0.110*** (0.012)	0.111*** (0.012)	0.111*** (0.012)	0.111*** (0.012)	0.111*** (0.012)
Q _{<i>i,t-1</i>}		0.019*** (0.002)	0.019*** (0.002)	0.019*** (0.002)	0.019*** (0.002)	0.019*** (0.002)	0.019*** (0.002)	0.019*** (0.002)	0.019*** (0.002)
Funding status _{<i>i,t-1</i>} / A _{<i>i,t-1</i>}				0.042* (0.024)	0.050*** (0.024)	0.026 (0.023)			
Underfunding _{<i>i,t-1</i>} / A _{<i>i,t-1</i>}							-0.164** (0.065)	-0.075 (0.066)	-0.040 (0.256)
Overfunding _{<i>i,t-1</i>} / A _{<i>i,t-1</i>}							0.020 (0.025)	0.021 (0.024)	0.148* (0.088)
Powers of funding variables	0	0	0	1	1	1	1	1	1, 2, 3
R ² (within)	0.098	0.098	0.100	0.100	0.101	0.101	0.100	0.101	0.101
Adjusted R ²	0.059	0.059	0.010	0.010	0.010	0.010	0.010	0.010	0.010
Alternate standard errors for mandatory contributions coefficient:									
Clustering by year			0.238			0.232		0.329	0.232
AR(1) model w/panel correlations			0.243			0.251		0.337	0.244

Standard errors are in parentheses. ***Significant at 1%; **significant at 5%; *significant at 10%.
All models contain firm-fixed effects and year-fixed effects. Standard errors in parentheses are heteroskedasticity-robust and clustered by firm.

Figure 5: Table II: Capital expenditures on pension and nonpension cash flows

Table III
Alternate Specifications for Regressions with Funding Status Controls

This table considers alternate specifications of the main test. The first four columns show results from two-stage least squares (2SLS) estimates of the model

$$\frac{I_{i,t}}{A_{i,t-1}} = \alpha_0 + \alpha_1 + \beta_1 Q_{i,t-1} + \beta_2 \frac{Y_{i,t}}{A_{i,t-1}} + \alpha_2 \gamma + \epsilon_{i,t}$$

$$\frac{Y_{i,t}}{A_{i,t-1}} = \alpha_0 + \alpha_1 + \beta_1 Q_{i,t-1} + \alpha_2 \gamma + \epsilon_{i,t}$$

where Z is mandatory contributions. Only coefficients from the investment equation are shown. In the first two columns, Y is total pension contributions and consists of nonpension cash flow and funding status itself. In the third and fourth columns, Y is total cash flow and x is pension funding status only. The firm fixed effects (α_1 and α_2) are only included where indicated. The remaining three columns test the robustness of the main specification in column (2c) for Table II to alternate specifications. RE is a random-effects model, and FD is a first-difference model. All models contain year-fixed effects.

Dependent Variable: Capital Expenditures _{i,t} / A _{i,t-1}						
<i>Variables instrumented with mandatory contributions:</i>						
Contributions (total) _{i,t} / A _{i,t-1}	-0.822*** (0.282)	-0.829*** (0.221)				
Cash flow _{i,t} / A _{i,t-1}			0.946* (0.518)	0.459*** (0.099)		
<i>Non-instrumented variables:</i>						
Contributions (mandatory) _{i,t} / A _{i,t-1}					-0.649** (0.282)	-0.863** (0.210)
Nonpension cash flow _{i,t} / A _{i,t-1}	0.115*** (0.009)	0.254*** (0.015)			0.099** (0.012)	0.145** (0.008)
Funding status _{i,t-1} / A _{i,t-1}	0.008 (0.021)	-0.060*** (0.016)	-0.080 (0.078)	-0.080*** (0.023)	0.018** (0.002)	0.013** (0.001)
Q _{i,t-1}	0.019*** (0.001)	0.000 (0.002)	-0.019 (0.023)	-0.019* (0.006)	0.034 (0.024)	0.003 (0.015)
Method	2SLS	2SLS	2SLS	2SLS	OLS	RE
Firm fixed effects	Y	N	Y	N	Y	N
Industry-year fixed effects	N	N	N	N	Y	N
Adjusted R ²	0.09	0.14	0.13	0.07	0.18	0.10
Observations	8,030	8,030	8,030	8,030	8,030	8,030
Firms	1,522	1,522	1,522	1,522	1,522	1,469

Standard errors are in parentheses. ***Significant at 1%; **significant at 5%; *significant at 10%.

Standard errors are heteroskedasticity-robust and clustered by firm except in the random effects specification.

Figure 6: Table III: Alternative specifications with funding status controls

5.5 Tables IV–V: other uses of funds and predictable vs. unexpected MCs

Read:

- Table IV finds no robust effect on R&D, but tobit specifications suggest reductions in acquisitions, dividends, and repurchases.
- Debt issuance, trade credit, and working capital do not rise enough to offset the pension cash drain.
- Table V separates mandatory contributions into predictable and unexpected parts.
- Unexpected MCs have negative coefficients from about -0.616 to -0.863.
- Predictable MCs are also negative in several specifications, though the interpretation is less clean.

Economic message. The investment effect is not simply replaced by other financing sources. Asset-market-driven pension shocks have a direct negative effect on investment.

Table IV
Research and Development, Acquisitions, and Financing Variables

This table estimates fixed-effects (FE) and Tobit models of the effect of required contributions on R&D, acquisitions, and financing variables. Each pair of rows shows the estimated coefficients and standard errors from the regression of an outcome variable on cash flow, Tobin's Q , the funding status of pension plans, and required contributions. All variables are scaled by assets and summary statistics of the dependent variable are presented in the right panel. Data on acquisitions, R&D, dividends, repurchases, debt, working capital, and trade credit are from Compustat.

Dependent Variable	Explanatory Variables				Model and Dependent Variable Statistics (Conditional Nonzero Statistics in Brackets)				
	Nonpension Cash Flow	Q _{i,t-1}	Funding Status	Mandatory Contributions	Model	Obs	Mean	Median	SD
R&D	-0.004 (0.006)	0.003*** (0.001)	0.013 (0.008)	0.087 (0.104)	FE	8,030 [3,588]	0.016 [0.036]	0.000 [0.022]	0.031 [0.038]
	-0.006 (0.008)	0.012*** (0.001)	0.087*** (0.012)	-0.002 (0.193)	Tobit				
Acquisitions	0.114*** (0.025)	0.011*** (0.004)	0.123** (0.051)	0.072 (0.450)	FE	8,030 [5,150]	0.025 [0.038]	0.000 [0.000]	0.075 [0.090]
	0.289*** (0.035)	0.010*** (0.004)	-0.157*** (0.049)	-4.846*** (0.952)	Tobit				
Dividends	0.015*** (0.003)	0.004*** (0.001)	0.018** (0.007)	-0.005 (0.061)	FE	8,030 [6,577]	0.018 [0.022]	0.013 [0.031]	0.020 [0.020]
	0.081*** (0.004)	0.010*** (0.000)	0.098*** (0.005)	-0.239** (0.098)	Tobit				
Repurchases	0.013* (0.008)	0.007*** (0.002)	0.027 (0.019)	-0.078 (0.219)	FE	8,030 [6,059]	0.011 [0.015]	0.000 [0.001]	0.026 [0.030]
	0.086*** (0.009)	0.011*** (0.001)	0.067*** (0.012)	-0.431* (0.227)	Tobit				
ΔDebt	-0.007 (0.073)	0.041*** (0.009)	0.390*** (0.118)	1.307 (1.566)	FE	8,030	0.053	0.023	0.170
Trade credit	-0.125*** (0.022)	-0.004 (0.003)	-0.080** (0.033)	-0.240 (0.428)	FE	8,030	-0.086	-0.071	0.117
Working capital	0.518*** (0.043)	0.008 (0.006)	0.118* (0.061)	-1.009 (0.750)	FE	7,310	0.192	0.172	0.199

Standard errors are in parentheses. ***Significant at 1%; **significant at 5%; *significant at 10%; †Contains industry and year fixed effects.

Standard errors are clustered by firm except in the Tobit models.

Figure 7: Table IV: R&D, acquisitions, and financing variables

Table V
“Predictable” vs. “Unexpected” Mandatory Contributions (MCs)

This table shows estimates of the response of capital expenditures to required pension contributions that have been separated into “predictable” and “unexpected” components. The fundamental computation applies the mandatory contributions function $M(\cdot)$ to expected values of the funding variables y and to the actual funding variables for each pension plan in the plan-level sample. The difference is unpredictable mandatory contributions. For the expected value of pension assets, an expected return is estimated based on the asset allocation of the pension fund (as implied by the actual return) and applied to lagged pension assets. Expected pension liabilities are equal to actual liabilities corrected for the change in the 30-year Treasury rate. See Section III.D for details. Pooled specifications contain nonlinear controls for the funding status (the first three powers). $\sigma(\text{Unexpected})$ is the standard deviation of the unexpected component, $\sigma(\text{Predictable})$ is the standard deviation of the predictable component, and $\rho(\text{Unexpected}, \text{Predictable})$ is the correlation between them. These calculations are done after removing firm- and year-fixed effects as called for by the regression model. All models contain year-fixed effects. The left panel contains firm-fixed effects. The right panel contains year effects and full nonlinear controls for the pension funding status.

	Fixed Effects Specifications			Pooled with Funding Controls		
Unexpected MCs/ $A_{i,t-1}$	-0.616*		-0.665**	-0.783**	-0.863**	
	(0.374)		(0.315)	(0.393)	(0.428)	
Predictable MCs/ $A_{i,t-1}$		-0.735*	-0.779**	-0.484	-0.593	
		(0.384)	(0.381)	(0.488)	(0.518)	
Nonpension cash flow/ $A_{i,t-1}$	0.111***	0.112***	0.111***	0.244***	0.244***	0.244***
	(0.012)	(0.012)	(0.012)	(0.015)	(0.015)	(0.015)
$Q_{i,t-1}$	0.019***	0.019***	0.019***	0.001	0.001	0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
$\sigma(\text{Unexpected})$			0.00117			0.00127
$\sigma(\text{Predictable})$			0.00115			0.00132
$\rho(\text{Unexpected}, \text{Predictable})$			-0.065			-0.121
Adjusted R^2	0.609	0.609	0.610	0.609	0.609	0.610
Observations	8,030	8,030	8,030	8,030	8,030	8,030

Standard errors are in parentheses. ***Significant at 1%; **significant at 5%; *significant at 10%. Standard errors are heteroskedasticity-robust and clustered by firm.

Figure 8: Table V: Predictable vs. unexpected mandatory contributions

5.6 Table VI and Figure 6: financing-constraint heterogeneity and timing

Read:

- Table VI shows stronger negative investment responses among younger firms, unrated/low-rated firms, low-dividend firms, low cash-minus-debt firms, and firms whose capital expenditures often exceed cash flow.
- The strongest coefficient is -1.523 for firms whose capital expenditures exceed cash flow more than one-third of the time.
- Figure 6 shows that investment drops in the contribution year and does not fully appear as a simple shift to adjacent years.

Economic message. The heterogeneity lines up with financing constraints: firms that are more likely to need external finance show larger investment cuts.

Table VI
Mandatory Contributions and Capital Expenditures
by Characteristics

This table reports results of regressions of the form

$$\frac{CAPX_{it}}{A_{i,t-1}} = \alpha_i + \alpha_t + \beta_1 Q_{i,t-1} + \beta_2 \frac{NonPensionCashFlow_{it}}{A_{i,t-1}} + \beta_3 \frac{MandatoryContributions_{it}}{A_{i,t-1}} + v_{it},$$

with the sample divided by hypothesized a priori indicators of financing constraints. A firm's age is defined as the number of years since its IPO year and is approximated as the number of years the firm is included in Compustat. The S&P Credit Rating is the S&P long-term domestic issuer credit rating (data280). The median dividend ratio is the within-firm median ratio of dividends to lagged book assets. The ratio of cash minus debt to assets is calculated as cash (data1) minus debt (data9 + data34) scaled by book assets, where debt consists of long-term debt plus debt in current liabilities.

Dependent Variable: CAPX _{<i>i,t</i>} / A _{<i>i,t-1</i>}				Explanatory Variables					
				Cash Flow		Q _{<i>i,t-1</i>}		Mandatory Contributions	
				Coeff	t-Stat	Coeff	t-Stat	Coeff	t-Stat
Count	Min	Max							
Panel 1: Sorting by Median Firm Age									
Age (youngest)	2,741	1	20	0.127	6.34	0.023	6.43	-0.954	-2.32
Age (middle)	2,790	21	34	0.095	4.93	0.019	4.44	-1.087	-2.15
Age (oldest)	2,499	35	48+	0.118	5.51	0.011	3.27	-0.578	-0.98
Panel 2: Sorting by Median S&P Credit Rating									
No S&P credit rating	3,597	-	-	0.090	5.95	0.019	5.37	-0.893	-2.30
S&P credit rating (low)	2,942	D	BBB+	0.118	5.82	0.025	5.89	-0.825	-1.77
S&P credit rating (high)	1,491	A-	AAA	0.214	5.38	0.011	3.38	0.639	0.50
Panel 3: Sorting by Median Dividend Ratio									
Low dividend	2,611	0.000	0.006	0.077	5.56	0.021	5.26	-1.136	-2.96
Middle dividend	2,611	0.006	0.023	0.160	5.64	0.029	5.61	0.086	0.19
High dividend	2,600	0.023	0.111	0.142	5.03	0.009	2.84	-0.156	-0.26
Panel 4: Sorting by Ratio of Cash Minus Debt to Assets									
Low cash minus debt	2,680	-9.852	-0.330	0.111	4.18	0.031	4.95	-1.682	-2.42
Middle cash minus debt	2,675	-0.330	-0.150	0.120	4.44	0.030	4.95	-0.863	-1.92
High Cash minus Debt	2,674	-0.150	0.795	0.115	5.94	0.013	4.31	-0.191	-0.64
Panel 5: Sorting by % of Firm Observations for Which CAPX > Cash Flow									
Never	2,905	0.000	0.000	0.215	10.01	0.006	2.87	-0.340	-1.12
Less than 1/3 of years	2,627	0.111	0.333	0.094	5.79	0.022	5.93	-0.420	-0.84
More than 1/3 of years	2,498	0.375	1.000	0.091	4.88	0.030	5.46	-1.523	-3.18

Standard errors are heteroskedasticity-robust and clustered by firm.

There is a question as to whether dividing the sample by age, credit rating,

(a) Table VI: Sorts by financing-constraint proxies

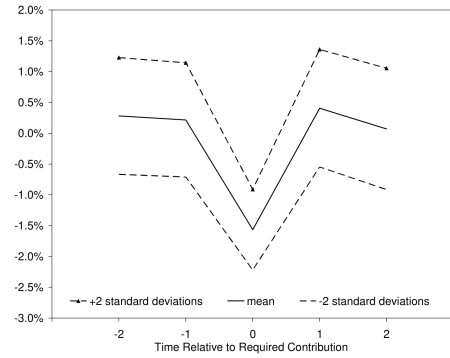


Figure 6. Investment around the time of large required pension contributions. This

(b) Figure 6: Timing around large contributions

5.7 Table VII: investment response of non-DB firms

Read:

- Table VII studies non-DB firms in industries where DB sponsors face mandatory contribution requirements.
- The industry pension requirement coefficient is positive in the all-non-DB-firm specification.
- The offset is strongest for non-DB firms that plausibly have available financial slack.
- Rauh interprets the non-DB offset as about 12 percent of the investment lost by pension sponsors.

Economic message. Some investment opportunities are reallocated to competitors, so the firm-level effect is larger than the aggregate effect.

Table VII
Investment Response of Non-DB Firms to Mandatory Contributions
of Pension Firms in the Same Industry

For firm i at time t in industry h , the following specification is estimated:

$$\frac{CAPX_{it}}{A_{i,t-1}} = \alpha_i + \alpha_t + \beta_1 Q_{i,t-1} + \beta_2 \frac{CashFlow_{it}}{A_{i,t-1}} + \beta_3 IndustryPensionRequirement_{h(i),t} + \varepsilon_{it}.$$

The analysis uses the 48-industry division of Fama and French (1997). The *Industry Pension Requirement* is constructed as the sum of estimated mandatory pension contributions at the industry-year level divided by the lagged balance sheet assets of nonpension assets in that industry-year cell. This variable is a measure of the magnitude of industry pension contributions relative to the size of the industry's nonpension firms. The results in this table report regressions on the sample of Compustat firms that do not have any defined benefit pension assets (data287 and data296 both equal zero). This is the sample in column (2.1) of Appendix Table I. The sorting is conducted on two plausible indicators of the firm's external financing requirement: the median cash ratio (the within-firm median of cash to balance sheet assets) and the share of firm observations for which capital expenditures are greater than cash inflows.

	Dependent Variable: $Capital\ Expenditures_{it}/A_{i,t-1}$						
	All Non-DB Firms	Sorted by Median Cash Ratio			Sorted by % of Firm Observations for Which Capital Expenditures > Cash Flow		
		Low	Middle	High	Low	Middle	High
Cash flow _{it} /A _{i,t-1}	0.075*** (0.010)	0.099*** (0.016)	0.080*** (0.015)	0.057*** (0.011)	0.141*** (0.015)	0.070*** (0.008)	0.066*** (0.018)
Q _{i,t-1}	0.005*** (0.001)	0.011*** (0.002)	0.010*** (0.002)	0.001 (0.001)	0.000 (0.001)	0.006*** (0.002)	0.006*** (0.002)
Industry pension requirement _{it}	0.073*** (0.015)	0.035 (0.066)	0.127*** (0.027)	0.061*** (0.020)	0.141*** (0.030)	0.078** (0.034)	0.001 (0.035)
Observations	46,848	15,569	15,581	15,581	15,681	15,779	15,283
Minimum value	—	0.000	0.043	0.195	0.000	0.286	0.714
Maximum value of sorting criterion	—	0.043	0.195	1.000	0.250	0.667	1.000

***Significant at 1%; **significant at 5%.
Standard errors are heteroskedasticity-robust and are clustered by industry.

contributions of zero (i.e., if the bracketed term in equation (11) were equal

Figure 10: Table VII: Non-DB firms in the same industry

6 Short summary

- The paper studies whether corporate investment depends on internal financial resources.
- It uses mandatory contributions to underfunded defined-benefit pension plans as cash-flow shocks.
- The contribution rules are legally determined and nonlinear in pension funding status.
- The main sample contains 8,030 firm-year observations from 1990–1998.
- Investment falls by roughly \$0.60–\$0.70 per \$1 of mandatory contributions in central specifications.
- The result survives controls for Tobin's Q, nonpension cash flow, firm fixed effects, year fixed effects, and pension funding status.
- The response is larger for firms that appear more financially constrained.
- The unexpected component of contributions, caused by asset-market and interest-rate movements, also reduces investment.
- Other financing sources do not fully offset the cash drain.
- Non-DB firms in the same industry take up part of the investment that DB sponsors forgo.

7 Conclusion

7.1 Core conclusion

Mandatory pension contributions reduce corporate investment. Because these contributions are tied to nonlinear pension funding rules rather than directly to investment opportunities, the evidence supports a causal role for internal financial resources in investment decisions.

7.2 Economic intuition

A firm with costly external finance cannot always replace an internal cash drain by borrowing or issuing equity. When pension law requires cash to be moved into the pension fund, constrained firms cut capital expenditures. The cuts are largest when outside finance is plausibly hardest to obtain.

7.3 Takeaway

Rauh provides large-sample evidence that financing constraints matter. The paper's contribution is not just another investment-cash flow regression; it uses pension funding rules to isolate cash-flow variation and shows that the marginal effect on investment is economically large.